CCC-P-91C
October 1, 1973
SUPERSEDING
Int. Fed. Spec. CCC-P-0091B(Army-GL)
March 11, 1969 and
Fed. Spec. CCC-P-91A
July 29, 1952

FEDERAL SPECIFICATION

PADDING; RUBBERIZED, CURLED HAIR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification is intended primarily to cover the requirements for supported rubberized hair padding for use as upholstered furniture cushioning and for other similar applications (see 6.1).

1.2 Classification.

1.2.1 Types and classes. The padding shall be of the following types and classes, as specified (see 6.2).

Type I - Curled hair (horsehair, cattlehair, and hog hair).
  Class 1 - 1 inch thick.
  Class 2 - 1-1/2 inches thick.
  Class 3 - 2 inches thick.

Type II - Hog hair.
  Class 4 - 5/8 inch thick.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-97 - Tape, Pressure Sensitive Adhesive, Filament Reinforced.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.
Fed. Test Method Std. No. 601 - Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified, before quantity production is commenced a finished sample 12 ± 1/2 inches by 12 ± 1/2 inches and the full thickness of the unit of the padding specified shall be submitted or made ready for approval of the contracting officer or his authorized representative for inspection to determine compliance with Section 4 of this specification. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification.

3.2 Materials.

3.2.1 Hair fibers. The curled animal hair fibers for padding shall consist of horsehair, cattlehair and hog hair. All hair fibers shall be of virgin quality and shall be well sterilized (see 4.3.5).

3.2.2 Cloth (supporting). The supporting or backing cloth shall conform to the following minimum requirements, when tested in accordance with 4.3.3:

Cloth - Cotton, rayon, polyester, or combination thereof.
Weave - Leno.
Yarns per inch:
  Warp - 12 (min.).
  Filling - 6 (min.).
Weight per square yd. - 2.5 ounces (min.).

3.2.3 Rubber. The rubber used for coating the hair fibers shall be either synthetic, natural, or combination thereof when identified in accordance with 4.3.4. No reclaimed rubber will be permitted.

3.2.3.1 Color. The color of the rubber shall be of any dark color that can be distinquished readily for the purpose of determining its penetration of the padding.

3.3 Padding (sheet).

3.3.1 Type I. Type I padding shall consist of 15 ± 5.0 percent of either horse-mane hair, horse-tail hair, cattle tail hair, or an intimate blend thereof, and 85 ± 5.0 percent prime winter hog hair (see 4.3.5). The hair fibers shall be coated with a minimum of 40 percent rubber by weight (see 3.3.7) and then suitably cured. The rubber shall be of the type specified in 3.2.3.

3.3.1.1 Thickness. Type I, class 1 padding shall be not less thab 1 inch thick. Type I, class 2 padding shall be not less than 1-1/2 inches thick. Type I, class 3 padding shall be not less than 2 inches thick. Tests shall be in accordance with 4.3.1.1.

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3.3.2 Type II. Type II padding shall consist of 100 percent prime winter hog hair (see 4.3.5) coated with a minimum of 40 percent rubber, by weight (see 3.3.7) and then suitably cured. The rubber shall be of the type specified in 3.2.3.

3.3.2.1 Thickness. The type II, class 4 padding shall be not less than 5/8 inch thick when tested in accordance with 4.3.1.1.

3.3.3 Cloth (supporting). The supporting or backing cloth, 3.2.2, shall be securely bonded to one side of the padding sheet when examined in accordance with 4.2.2.1.

3.3.4 Rubber coating. Padding shall contain not less than 40.0 percent, by weight, of rubber 3.2.3, thoroughly coating the hair and suitably cured, when tested in accordance with 4.3.4.1.

3.3.5 Tackiness. There shall be no noticeable tackiness of the padding when examined in accordance with 4.2.2.1.

3.3.6 Compression set. The compression set of the padding shall not exceed 7.0 percent of the original thickness of the padding when tested in accordance with 4.3.1.1.2.

3.3.7 Weight. The finished weight of the paddings shall be as follows, when tested in accordance with 4.3.2:

5/8 inch thick - 2.5 ounces per square foot (min.).
1 inch thick - 3.5 ounces per square foot (min.).
1-1/2 inches thick - 4.0 ounces per square foot (min.).
2 inches thick - 5.0 ounces per square foot (min.).

3.3.8 Dimensions of padding sheets. Unless otherwise specified the padding shall measure 24 inches by 72 inches with a plus or minus tolerance of 1/2 inch in either or both directions (see 4.2.2.2).

3.3.9 Workmanship. The finished padding shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL's).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchaser order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in the specification or applicable purchase document.

4.2.2 Inspection of end item.

4.2.2.1 Visual examination. Examination of the end item shall be in accordance with the defects listed below and the acceptable quality levels (AQL's) shall be as specified in 4.2.2.3. For purposes of sampling, all padding of the same type, class and dimensions offered for inspection at one time shall be considered a lot. The sample unit shall be one completely finished sheet of padding.

Examine	Defects
Padding	Not uniform. Contains foreign matter. Binder tacky. Color not dark. Any delamination of support cloth.

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4.2.2.2 Dimensional examination. Inspection shall be made of the finished padding to determine compliance with the dimensions as specified. Any dimensions not within specified tolerances shall be classified as a defect.

4.2.2.3 Inspection levels and acceptable quality levels (AQL's) for examination. The inspection levels for determining sample size and the acceptable quality levels (AQL's) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1	S-1	1.5
4.2.2.2	S-1	4.0

4.2.3 Examination of preparation for delivery requirements. When procurement is made by the Government an examination shall be made to determine compliance with the packaging, packing and marking requirements of Section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, incorrect location, or method of application.
Materials	Any nonconforming component, component missing, damaged, or otherwise defective, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulged or distorted container.
Contents	Number of items per container is more or less than required. Not packed as specified.

4.2.4 Sampling for tests. The completely finished sheet of padding shall be tested for the characteristics listed in 4.3 through 4.3.5. The test requirements shall be applicable to the individual unit. For purposes of sampling, all sheets of padding offered for inspection at one time shall constitute a lot. The sample unit shall be two completely finished sheets of padding. The sample size shall be 5. There shall be no evidence of failure of any sample unit to meet the requirement specified.

4.3 Tests.

4.3.1 Test conditions. Unless otherwise specified hereinafter all physical testing shall be made under temperature and humidity conditions as specified in Section 7 of Fed. Std. No. 601.

4.3.1.1 Test specimens. Unless otherwise specified, the test specimens shall be pieces 4 inches wide by 4 inches long, cut from the sample selected. The specimens shall be tested in the thickness as received, except that specimens having a nominal thickness of less than 2 inches shall be tested in combination, as required, to provide a minimum test thickness of 2 inches.

4.3.1.1.1 Initial thickness. The test specimen shall be placed upon a horizontal plane (flat) surface and on top of it, properly centered, shall be placed a suitable 5 by 5 inch flat piece of metal, plywood, or plastic. Upon this plate shall be placed weight that will apply to the specimen a total compressive load of 4 pounds (1/4 pound per square inch of specimen area). The initial thickness recorded shall be the average of 3 determinations of the height measurements taken at the four corners of the 5 by 5 inch plate, holding the required test load.

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4.3.1.1.2 Compression set. The specimen shall be a 4-inch square. In no case shall either contact dimension be less than the thickness of the specimen. The compression device shall consist of two flat plates arranged parallel and supported by threaded bolts in such a manner that the space between the plates can be adjusted to give the required compression of the specimen. The percentage compression of the padding is controlled by means of spacers placed between the compression plates on each side of the specimen. The plates shall be of sufficient thickness to withstand the required compression stresses without bending. The surfaces contacting the specimen shall be polished and shall be thoroughly cleaned before using. The specimen shall be measured for original thickness and recorded as T_1 placed between the plates of the compression device and compressed to 50 percent of the original thickness, 4.3.1.1.1, by tightening the bolts so that the plates are drawn together uniformly until they contact the spacers. For type I padding a weight of 17-1/2 ± 1 pounds shall be used for compression. For type II padding a minimum weight of 13-1/2 pounds and maximum weight of 20 pounds shall be used for compression. The assembled compression apparatus and specimen shall be maintained for 3 minutes. At the end of the test period, the specimen shall be removed from the apparatus and allowed to rest on a wooden surface at room temperature for 2 minutes. At the end of the two minute period, the thickness of the specimen shall be measured again and the result recorded as T_2. The compression set of the specimen shall be calculated as follows:

$$\text{Compression set, percent} = \frac{T_1 - T_2}{T_1} \times 100$$

where: T_1 = the original thickness of the specimen

T_2 = the thickness of the specimen 2 minutes after removal from the compressor.

4.3.2 Weight. The finished weight of the padding shall be tested in accordance with method 5041 of Fed. Std. No. 191.

4.3.3 Cloth (supporting). The supporting cloth for padding shall be tested as a component item, in accordance with the following methods of Fed. Std. No. 191; and as otherwise indicated:

Characteristic	Test method
Fiber identification (cloth)	1200, 1500, 1600
Weave	Visual
Yarns per inch	5050
Weight per square yard	5041

4.3.4 Rubber. The rubber, as a component, shall be identified in accordance with method 15011 of Fed. Std. No. 601. When specified, a certificate of compliance will be acceptable for the stated requirement in 3.2.3.

4.3.4.1 Rubber content. A certificate of compliance shall be required from the supplier stating the rubber content of the padding meets the requirements of paragraph 3.3.4.

4.3.5 Hair fibers. A certificate of compliance shall be required from the supplier stating the fiber content of the padding and verification that hair fibers are of virgin quality and have been sterilized (see 3.2.1, 3.3.1 and 3.3.2).

5. PREPARATION FOR DELIVERY

5.1 Packing. When procurements are made by the Government packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Padding shall be packed in a snug-fitting fiberboard shipping container conforming to style optional, grade V2s of PPP-B-636, except that dimensional limitations shall be waved. Closure, waterproofing by means of tape, and reinforcing with pressure-sensitive adhesive tape conforming to PPP-T-97 shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level B. Padding shall be packed in a snug-fitting fiberboard shipping container conforming to style optional, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636, except that dimensional limitations shall be waived. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636.

5.1.2.1 When specified, the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto. The shipping container material may also be grade V4s of PPP-F-320.

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5.1.3 Level C. Padding shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I, classes 2 and 3 shall be used primarily for back padding on upholstered furniture. Type I, class 1 shall be used primarily for arm padding and general upholstery work of furniture. Type II, class 4 shall be used primarily for platform decking on upholstered furniture.

6.2 Ordering data. Purchasers should select the perferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Types and classes (1.2.1).
c. Whether preproduction sample is required (3.1).
d. Packing (5.1).

PREPARING ACTIVITY:

GSA-FSS

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Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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